

Robotique et Apprentissage

Introduction

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Organization

- ▶ Evaluation based on all labs. 3 notes : OS labs, NB lab, SD labs
- ▶ Change students pairs each time
- ▶ If Covid restrictions (?): labs at home, teacher available through visio tool, discord-like between students
- ▶ 5 identical sections:
 - ▶ Week 1 (friday 16-18): class,
 - ▶ Week 2 (thursday 13:45-18): GR1 lab session,
 - ▶ Week 3 (thursday 13:45-18): GR2 lab session.

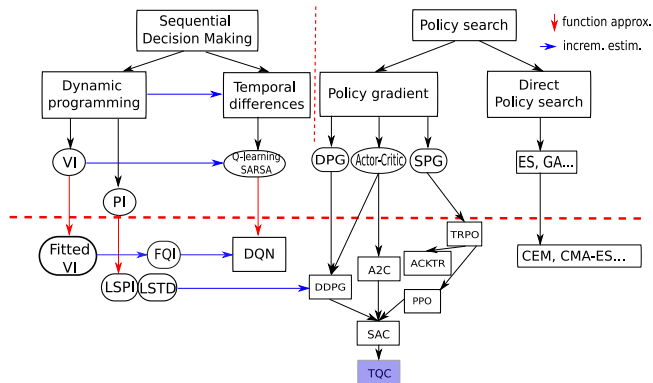
Schedule of lessons (2022)

- ▶ Sigaud, 28/1: Regression in robotics: learning mechanical models of robots
- ▶ Regression lab, 3/2 ; 10/2
- ▶ Sigaud, 11/2: Reinforcement learning: the basics
- ▶ Reinforcement learning lab, 17/2 ; 24/2
- ▶ Bredeche, 25/2: Evolution strategies (**one week off**)
- ▶ ES and Robotics lab, 10/3 ; (**one week off**) 24/3
- ▶ Doncieux, 25/3: Multi-Objective Optimization with evolutionary methods
- ▶ X-Obj Evo lab, 31/3 ; deadline 7/4
- ▶ Doncieux, 8/4: Structured representations: From genetic programming to neuro-evolution
- ▶ Structured rep-Evo lab, 14/4, 21/4

Outline of the regression part

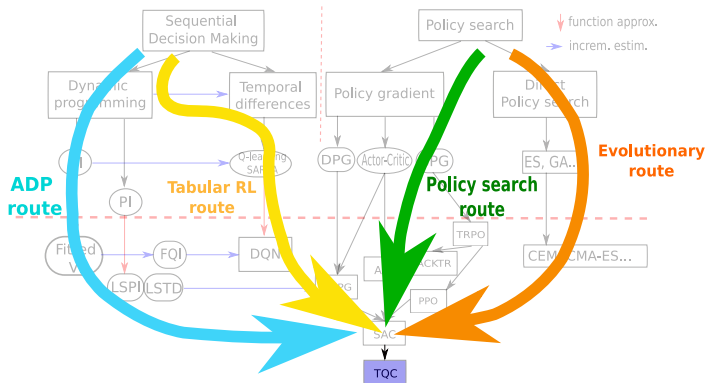
- ▶ Linear least quares
- ▶ Locally Weighted Regression
- ▶ Batch projection methods
- ▶ Gradient Descent methods
- ▶ Applications to RBFNs
- ▶ Applications to Neural Networks

Outline of the policy search part



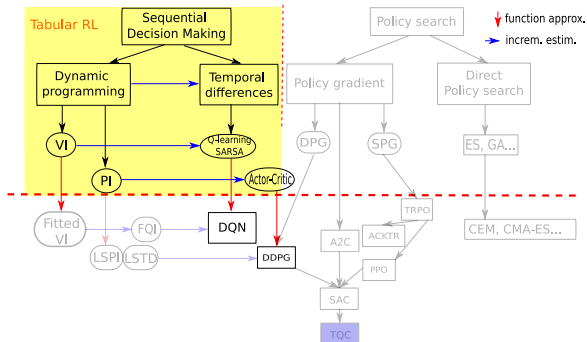
- A very partial view of the whole RL literature

The four routes to deep RL



- Four different ways to come to Deep RL

The Tabular RL route

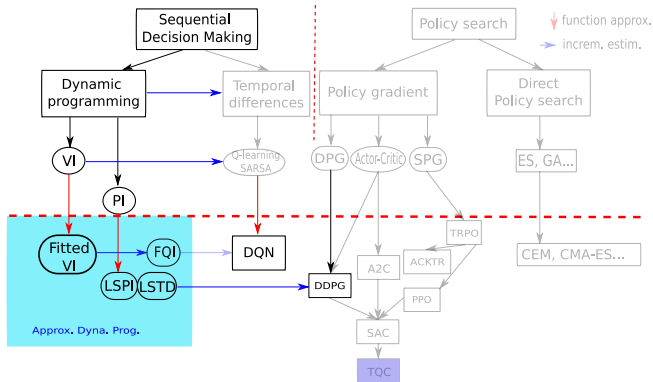


- ▶ The favorite route of beginners
- ▶ Start from Sutton&Barto, present Q-learning, SARSA and Actor-Critic
- ▶ Add function approximation with NNs, go to DQN, then DDPG
- ▶ We will start this route together



Sutton, R. S. & Barto, A. G. (1998) *Reinforcement Learning: An Introduction*. MIT Press.

The Approximate Dynamic Programming route

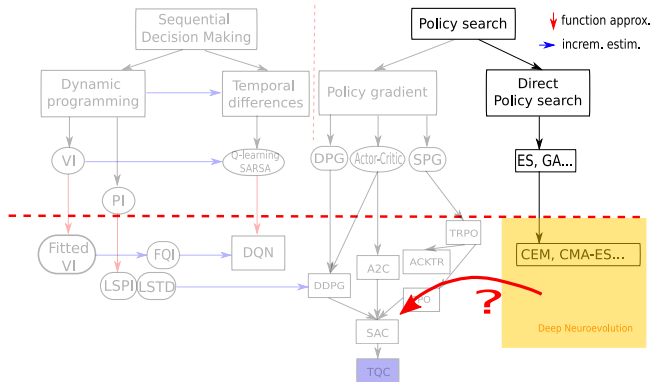


- ▶ The favorite route of mathematicians
- ▶ I never travelled this route



Warren B. Powell. *Approximate Dynamic Programming: Solving the curses of dimensionality*, volume 703. John Wiley & Sons, 2007

The Evolutionary route

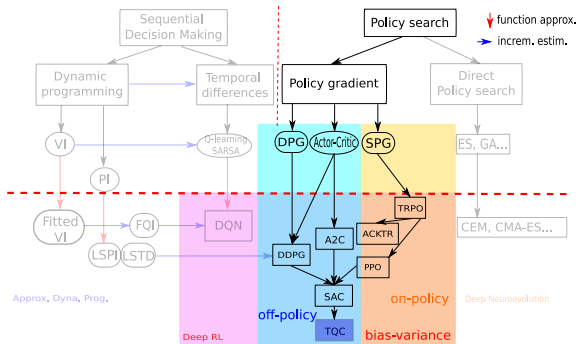


- ▶ Much more efficient than RL people think
- ▶ Studied in NB and SD's lessons



Tim Salimans, Jonathan Ho, Xi Chen, and Ilya Sutskever. Evolution strategies as a scalable alternative to reinforcement learning. *arXiv preprint arXiv:1703.03864*, 2017.

The Policy Search route



- ▶ The favorite route of roboticists
- ▶ Studied next year



Marc P. Deisenroth, Gerhard Neumann, Jan Peters, et al. A survey on policy search for robotics. *Foundations and Trends® in Robotics*, 2(1-2):1-142, 2013

Online Machine Learning Classes

- ▶ My Channel:
<https://www.youtube.com/OlivierSigaud>
- ▶ Three available playlists
 - ▶ Regression class: 6 videos, 1h20
<https://www.youtube.com/watch?v=JxUyMs6os3Q&list=PLe5mY-Da-ksURGcelU-egwtWZXxVgZ76a>
 - ▶ Reinforcement learning class: 8 videos, 2h40
<https://www.youtube.com/watch?v=9gzL3QQzvQ4&list=PLe5mY-Da-ksWV330WbfazLUyOuR59sers>
 - ▶ Deep Policy Search class: 7 videos, 4h
<https://www.youtube.com/watch?v=XwkN64MnZJQ&list=PLe5mY-Da-ksVku6MoVI5DRLES529BheYI>

Any question?



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